

Optical SMOKE ALARM

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This optical smoke alarm (Fig. 1) is built around opto-interrupter MOC7811 (IC1) and dual op-amp LM358 (IC2).

When thick black smoke enters in the gap (slotted area) of the opto-interrupter sensor (IC1), its internal photo transistor stops conduction. As a result, output of the dual op-amp goes low. So transistor T1 stops conduction and T2 starts conducting. This makes the piezo buzzer (PZ1) sound an alarm and LED1 to glow.

Potmeter VR1 is used to control intensity of the internal LED of MOC7811. Potmeter VR2 is used to control the reference voltage at pin 3 of IC2.

PARTS LIST

Semiconductors:

- | | |
|--------|-----------------------------------|
| IC1 | - MOC7811 opto-interrupter module |
| IC2 | - LM358 dual op-amp |
| T1, T2 | - BC547 npn transistor |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|--------|-------------------------|
| R1 | - 100-ohm |
| R2 | - 33-kilo-ohm |
| R3, R4 | - 10-kilo-ohm |
| R5 | - 680-ohm |
| VR1 | - 4.7-kilo-ohm potmeter |
| VR2 | - 10-kilo-ohm potmeter |

Capacitors:

- C1 - 100 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin terminal connector
PZ1 - Piezo buzzer
- 4-pin connector for IC1

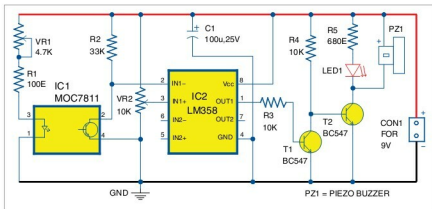


Fig. 1: Circuit diagram of the optical smoke alarm

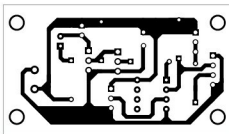


Fig. 2: An actual-size PCB pattern of the optical smoke alarm

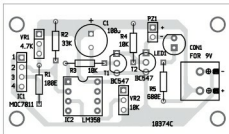


Fig. 3: Component layout of the PCB

The circuit works off a 9V battery.

Construction and testing

An actual-size, single-side PCB for the optical smoke alarm is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable box. Install MOC7811 at a proper location for monitoring smoke or fire.

Fix CON1, LED1 and piezo buzzer on the front side of the cabinet. VR2 is used for reference setting. Connect a 9V battery to CON1. **EPY**



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